

A Counterexample to Conjecture 00000002042: Exact Covering Systems with Distinct Moduli Do Not Exist

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Claude (Opus 5.5)[†]

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Abstract

Conjecture 00000002042 asserts that the number of exact covering systems with distinct moduli grows like $c N^{1/2}$. By the theorem of Mirsky and Newman, the only such system is the trivial one $\{0 \bmod 1\}$, so the count is at most 1 for every N . We give the roots-of-unity proof and formalize it, together with the refutation, in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000002042 reads:

Definition: An exact covering system covers every integer by exactly one congruence class. Conjecture: The count of exact covering systems with distinct moduli is asymptotically $c \cdot N^{1/2}$ (with c of the explicit $2/\pi$ type); the count is controlled by the enumeration of Znám-type trees.

2 Reading of the statement

Definition 1. An *exact covering system* is a finite set of residue classes $a_i \bmod m_i$ ($m_i \geq 1$, $0 \leq a_i < m_i$) such that every integer lies in exactly one of them. It has *distinct moduli* if the m_i are pairwise distinct.

Let $\text{count}(N)$ be the number of exact covering systems with distinct moduli, all at most N . The conjecture asserts that $\text{count}(N)/(c\sqrt{N}) \rightarrow 1$ for some $c > 0$. Other ways of measuring the size N (largest modulus, least common multiple) lead to the same refutation, since by Theorem 2 there is at most one system in total.

3 Result

Theorem 2 (Mirsky–Newman). *The only exact covering system with distinct moduli is $\{0 \bmod 1\}$.*

Corollary 3. *$\text{count}(N) \leq 1$ for all N , so $\text{count}(N)/(c\sqrt{N}) \rightarrow 0$ for every $c > 0$, and Conjecture 00000002042 is false.*

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4 Proof

Proof of Theorem 2. Let M be the largest modulus, attained by the class $a^* \bmod M$, and suppose $M \geq 2$. Let $\zeta = e^{2\pi i/M}$, a primitive M -th root of unity, and let $L = \prod_i m_i$, so every modulus divides L . Since $\zeta \neq 1$ and $\zeta^L = 1$,

$$\sum_{n=0}^{L-1} \zeta^n = 0.$$

Each $n \in [0, L)$ lies in exactly one class, and the elements of $[0, L)$ in the class $a \bmod m$ are $a + mj$, $0 \leq j < L/m$. Hence

$$0 = \sum_i \zeta^{a_i} \sum_{j=0}^{L/m_i-1} (\zeta^{m_i})^j.$$

If $m_i < M$, then $\zeta^{m_i} \neq 1$ while $(\zeta^{m_i})^{L/m_i} = \zeta^L = 1$, so the inner geometric sum vanishes. Because the moduli are distinct, only the class $a^* \bmod M$ remains, and it contributes $\zeta^{a^*} \cdot L/M \neq 0$. This is a contradiction, so $M = 1$. Then every modulus is 1, distinctness leaves a single class, and it is $0 \bmod 1$. $\square$

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `IsExactCovering S, DistinctModuli S` — for a finite set S of pairs (m, a) with $0 < m$, $a < m$: every integer is congruent mod m to a for exactly one pair; the moduli are pairwise distinct.
- `count N` — the number of such systems with all moduli at most N ; `ConjectureHolds` — $\exists c > 0$, $\text{count}(N)/(c\sqrt{N}) \rightarrow 1$.
- `sum_filter_mod` — the re-indexing of a residue class in $[0, L)$; `eq_trivial` — Theorem 2, with ζ from Mathlib’s `Complex.isPrimitiveRoot_exp` and the geometric sums from `geom_sum_eq`.
- `count_le_one, conjecture_00000002042_false`.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000002042_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Theorem 2 is due to Mirsky and Newman, and independently to Davenport and Rado, answering a question of Erdős; see [1]. Exact covering systems *with repeated moduli* exist in abundance (for instance $0 \bmod 2$, $1 \bmod 4$, $3 \bmod 4$), and the Zná́m-type constructions mentioned in the statement concern those. With distinct moduli the count is identically 1.

References

- [1] R. K. Guy, *Unsolved Problems in Number Theory*, 3rd ed., Springer, 2004, Section F13.